# Sustainability of Embedded Systems

Team 05

## WHAT DO WE NOW KNOW THAT WE DID NOT KNOW BEFORE? 01

**Functionally identical code can differ significantly in energy use.**

- A plain interrupt drew more power than busy-wait.
- Context switching and the interrupt-controller logic offset the gain.
- Interrupts only win when combined with a CPU sleep mode (Wait-For-Interrupt).

## ARE THE REPORTED CONCLUSIONS GENERALISABLE? 02

**Only with reservations.**

- Specific to the STM32F429 / CT-Board and a low-frequency event.
- Limited signal-to-noise ratio.
- A negative current offset is visible in the readings.
- The hardware-timer variant needed over 150 lines of code for a near-identical result → poor effort-to-gain ratio.

## IN WHICH WAY CAN WE RELATE TO THE LESSONS LEARNED? 03

**The simplest implementation is rarely the most efficient one.**

- Reducing active processor time and unintended input/output (e.g. logging) outweighs local code tweaks.
- A forgotten debug output caused a roughly hundred-fold slowdown.
- Software design (not hardware) determines long-term efficiency.

## WHY IS THE EXPERIENCE NOT VERY REPRESENTATIVE? 04

**The CT-Board is a dev-platform, not an IoT node.**

- 12 V input vs. 1.5–3.3 V on battery-powered Internet-of-Things devices.
- Onboard LEDs and auxiliary components draw up to two hundred times the processor's own baseline current.
- Measurement noise prevents transfer of absolute values to production hardware.